

Airborne Hazards and Human Health

Lecture 2



Case Study 1: London Smog, 1952



A man guides a London bus through thick fog with a flaming torch during the 1952 Great Smog



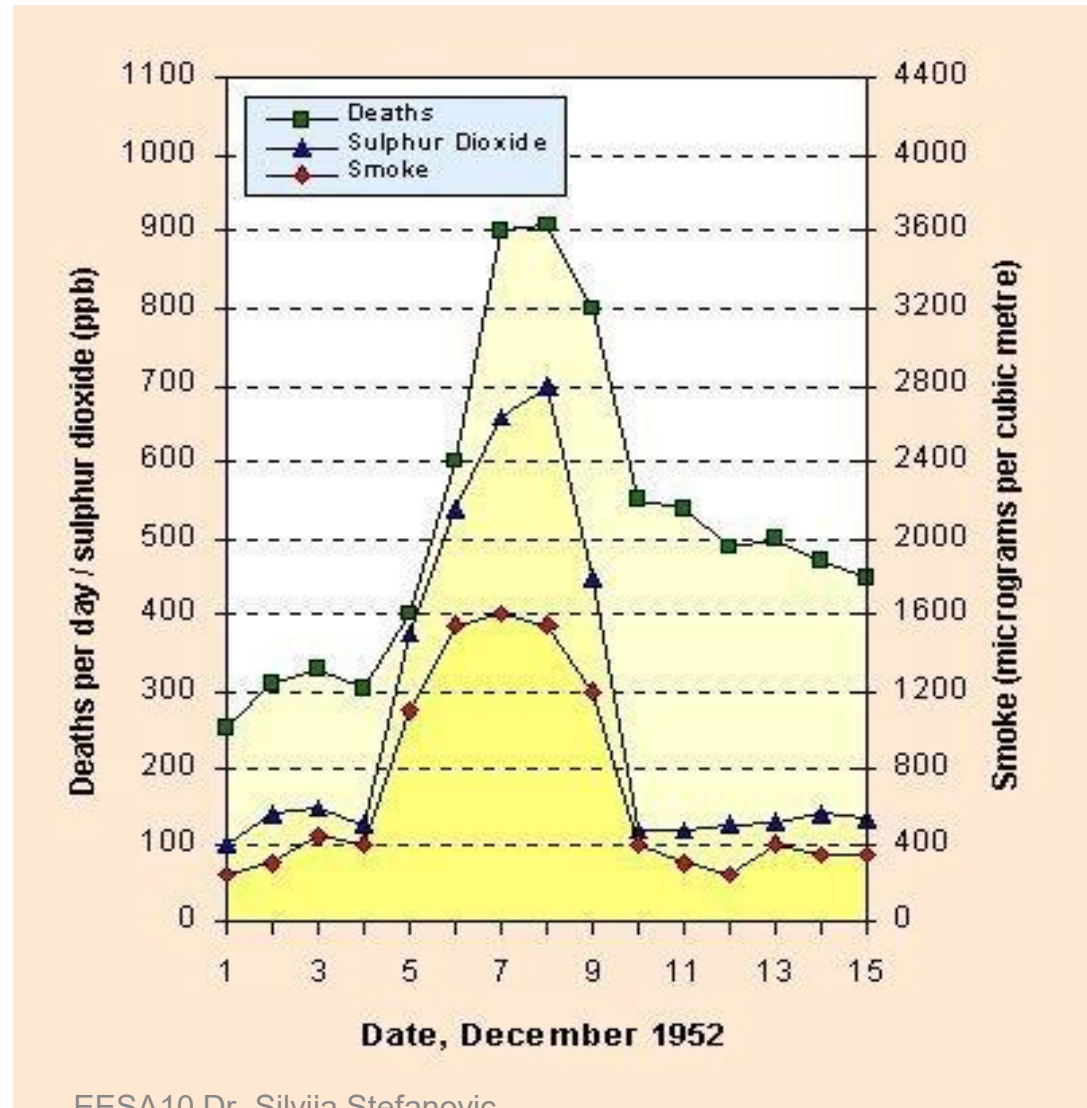
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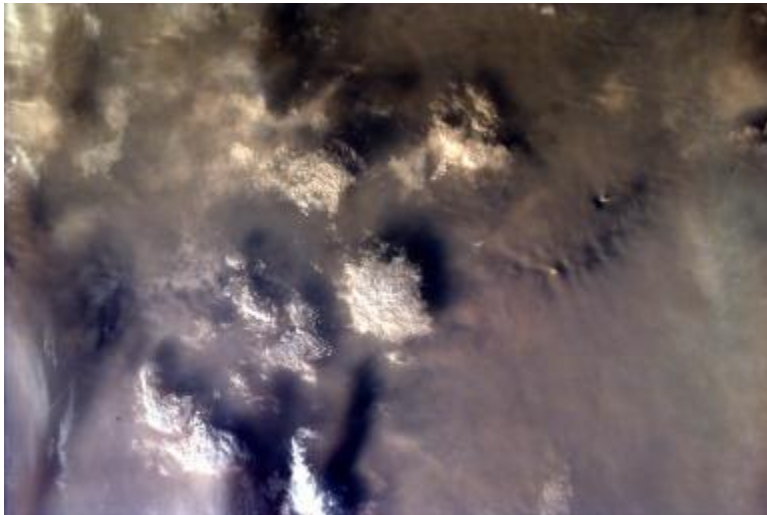
The smog was the result of coal burning

The Relationship Between Smoke And Sulfur Dioxide Pollution And Deaths During The Great London Smog, December 1952, Source: Wilkins, 1954

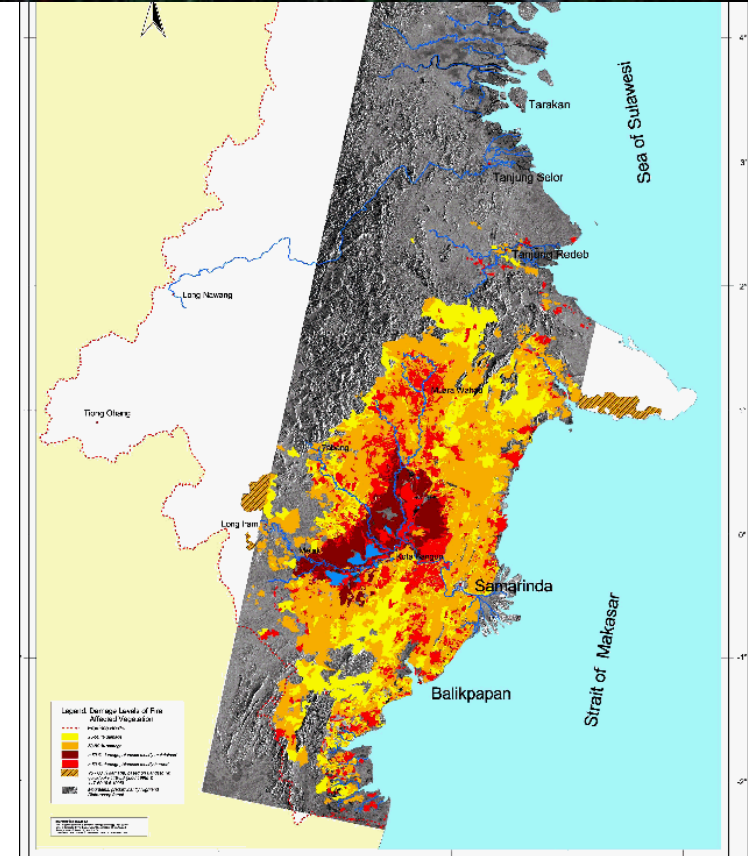
Figure shows the average smoke and sulphur dioxide levels for 12 London sites and the relationship with deaths recorded during the smog period in December 1952. The peak in the number of deaths coincided with the peak in both smoke and sulphur dioxide pollution levels.



Case Study 2: Indonesian Fires, 1997



Smoke over Lake Toba, Sumatra, Indonesia, photographed by the Space Shuttle *Atlantis*. Smoke from fires set to clear land for agriculture in Indonesia at one time in 1997 blanketed an area larger than the continental United States.



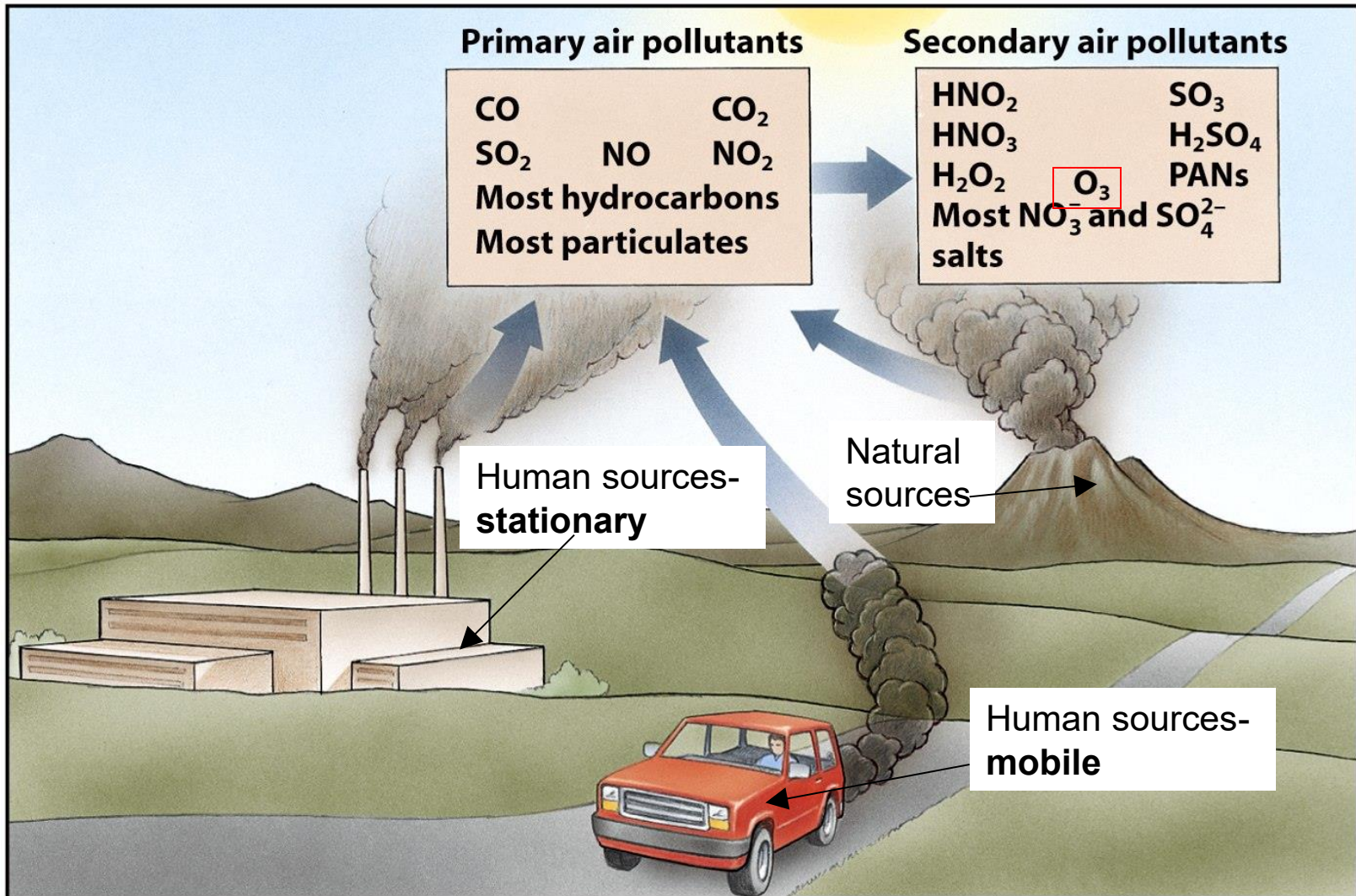
Fire damage classification of the 1997-1998 fires in East Kalimantan, Indonesia, based on ERS-SAR images.

Airborne Hazards

- Outdoor air pollution
- Indoor air pollution



Sources of Outdoor Air Pollution



Air Pollution from Burning Fossil Fuels

Table 4.1 Key Sources of Major Air Pollutants from the Burning of Fossil Fuels

Pollutant	<i>Sources of Pollutants</i>			
	Vehicles (gasoline, diesel)	Electric Power Plants (coal, oil)	Heating Buildings (oil, natural gas)	Manufacturing (coal, oil, natural gas)
<i>Basic products of the combustion process</i>				
Carbon dioxide (CO ₂)	✓	✓	✓	✓
Carbon monoxide (CO)	✓			
Nitrous oxide (N ₂ O)	✓	✓		
Nitrogen dioxide (NO ₂), Nitric oxide (NO)	✓	✓		
Sulfur dioxide (SO ₂)		✓		(some)
Particulate matter (PM)	✓	✓	✓	✓

Air Pollution from Burning Fossil Fuels

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<i>Other pollutants liberated by combustion</i>				
Mercury (from coal)		✓		(some)
Lead (from leaded gasoline)	✓			
Volatile organic compounds (from gasoline)	✓			

Health Effects of Outdoor Air Pollution

- The effects depend on the dose or concentration

Asthma

- Particulates and/or SO₂ can irritate bronchial passages leading to severe difficulties in breathing
- Prevalence in the US increased 34%
- The incidence among a children in Australia was one in five, a doubling of the rate in less than 20 years
- Indoor air pollution is also significant cause

Chronic bronchitis

- Occurs when an excessive amount of mucus is produced in bronchi which results in a lasting cough
- SO₂ and smoking is related to Chronic bronchitis

Health Effects of Outdoor Air Pollution

Pulmonary emphysema

- Weakening of the wall of alveoli, they become enlarged and loss their resilience
- Shortness of breath is the primary symptom
- NO₂ is related to emphysema

–Lung Cancer
–Heart disease
–Toxic poisoning
–Eye irritation
–Birth defects

} Caused or exacerbated by exposure to air pollution

Seven Common Outdoor Air Pollutants

- **Primary air pollutants**
 - Particulate matter
 - Carbon monoxide
 - Nitrogen oxides
 - Sulphur oxides
 - VOC (Volatile Organic Compounds)
 - Lead
- **Secondary air pollutant**
 - Ground level Ozone



Early Morning Fog Shots in Toronto

Particulate Matter

- Particles found in the air (dust, soot, smoke, and liquid droplets)
- Vehicles, factories, construction sites, tilled fields, stone crashing, and all kinds of burning
- Some formed in the air
- Serious health effects
- Big and small

Particulate Matter

- Particulates classified by size
 - PM₁₀ —respirable
 - PM_{2.5} —“fine” (mostly from combustion)
 - Ultrafine particulates

Table 4.3 Likely Sources and Fates of Respirable Particulates, by Size Category

Diameter (microns)	Key Sources	Penetration and Fate in Body
2.5 to 10	Natural and mechanical sources	Settle out in trachea and bronchi; are removed via mucociliary escalator
0.1 to < 2.5	Combustion	Reach small airways and alveoli; in alveoli, are removed by macrophages
< 0.1 (ultrafine)	Combustion (especially of diesel fuel)	Can pass through alveolar wall into bloodstream

Carbon Monoxide

- Odourless, colourless gas
- Incomplete burning of carbon containing fuels
- Heaters, woodstoves, gas stoves, fireplaces, water heaters, automobile exhaust, and tobacco smoke
- Bigger problem for indoor air pollution

Carbon Monoxide

- 1 000 people die each year in US as result of CO poisoning (indoor air)
- Sometimes confused with flu or food poisoning
- Fetuses, infants, elderly and people with heart and respiratory illnesses are at high risk for adverse health effects

Health Effects of Carbon Monoxide

- Interferes with the delivery of oxygen in the blood to the rest of the body
- Worsen cardiovascular conditions
- Fatigue
- Headache
- Weakness
- Confusion
- Disorientation, loss of coordination
- Nausea, Dizziness, Death



Carbon Monoxide
Poisoning

Prevention

- Never leave a car engine running in a shed or garage or in any enclosed space
- Proper selection, installation, and maintenance of appliances
- Correct use of appliances
- Good ventilation
- Use CO detectors



Nitrogen Oxides (NO_x)

- Formed in any type of combustion process
- Involved in formation of ground level ozone
- Forms nitrate particles, and acid aerosols
- Contribute to formation of acid rain
- Transported over long distances

Sulphur Oxides (SO_x)

- Burning of coal and oil, extraction of metals from ore
- SO₂ dissolve in water vapour to form acids
- Acids react with other gases and particles and form sulphates
- Transported over long distances
- Respiratory illnesses, aggravates existing heart and lung diseases

VOC (Volatile Organic Compounds)

- Variety of organic compounds used as solvent in industry, automobiles
- Hydrocarbons (HC): methane, butane, propane
- Some produce photochemical smog (will be explained later)

Lead (Pb)

- Metal
- Vehicles and industrial sources
- Leaded gasoline
- Deposit on soil and water
- Children accidentally can eat soil
- Particularly affects young children

Lead (Pb)

- Lead is neurotoxicant
 - Effects on
 - IQ, cognitive & neurological performance more broadly
 - Cardiovascular mortality, increased lung cancer risk, kidney toxicity, high blood pressure, reproductive effects, and oral health impacts
 - Heavier burden of exposure on poor, nonwhite populations

Ozone (O₃)

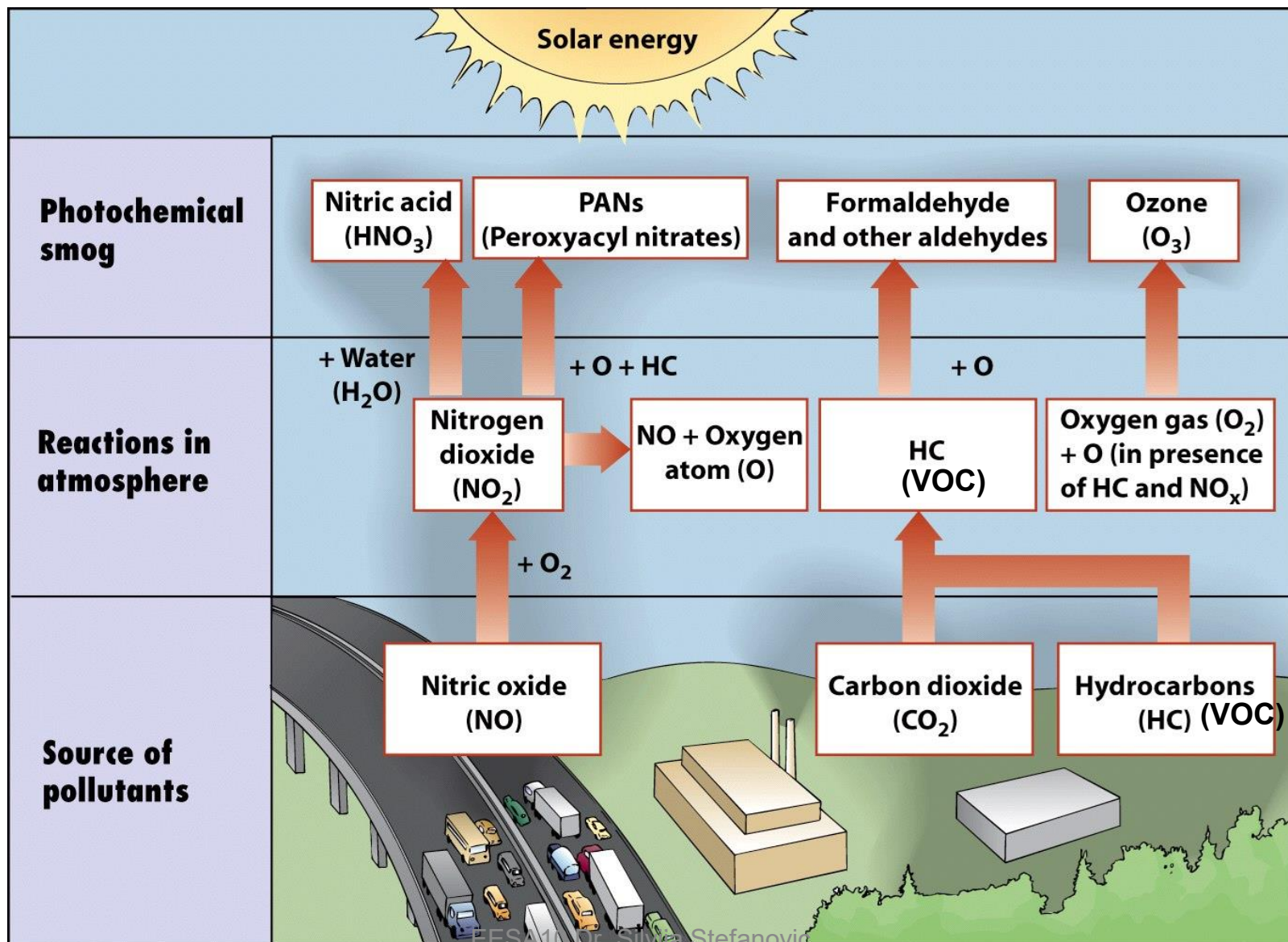
VOC + NO_x + Heat + Sunlight = **Ozone**

- Summertime pollutant, time of the day, seasonal, climate
- Good in stratosphere
- Bad on a ground
- Transported on long distances
- Lung damage (small airways)
- Shortness of breath, chest tightness, coughing and nausea
- Irritate and damaged eyes, nose, sinuses and throat
- Problem for people who exercise outdoors during the concentration picks

Smog

- **S**moke + **f**og = **S**mo**g** (first used in 1905 in England)
- Sulphurous smog- *London type of smog or industrial smog*
- Photochemical smog - *L.A. type of smog or brown air*

Formation of Photochemical Smog



Airborne Hazards

- Outdoor air pollution
- Indoor air pollution



Indoor Air Pollution

- Contains 2-5 times higher concentration of hazardous pollutants than outdoor air
- Buildings more airtight to conserve energy, inadequate ventilation
- People spend about 90% of their time indoors
- Children, pregnant women, elderly, and people with chronic illnesses are more sensitive

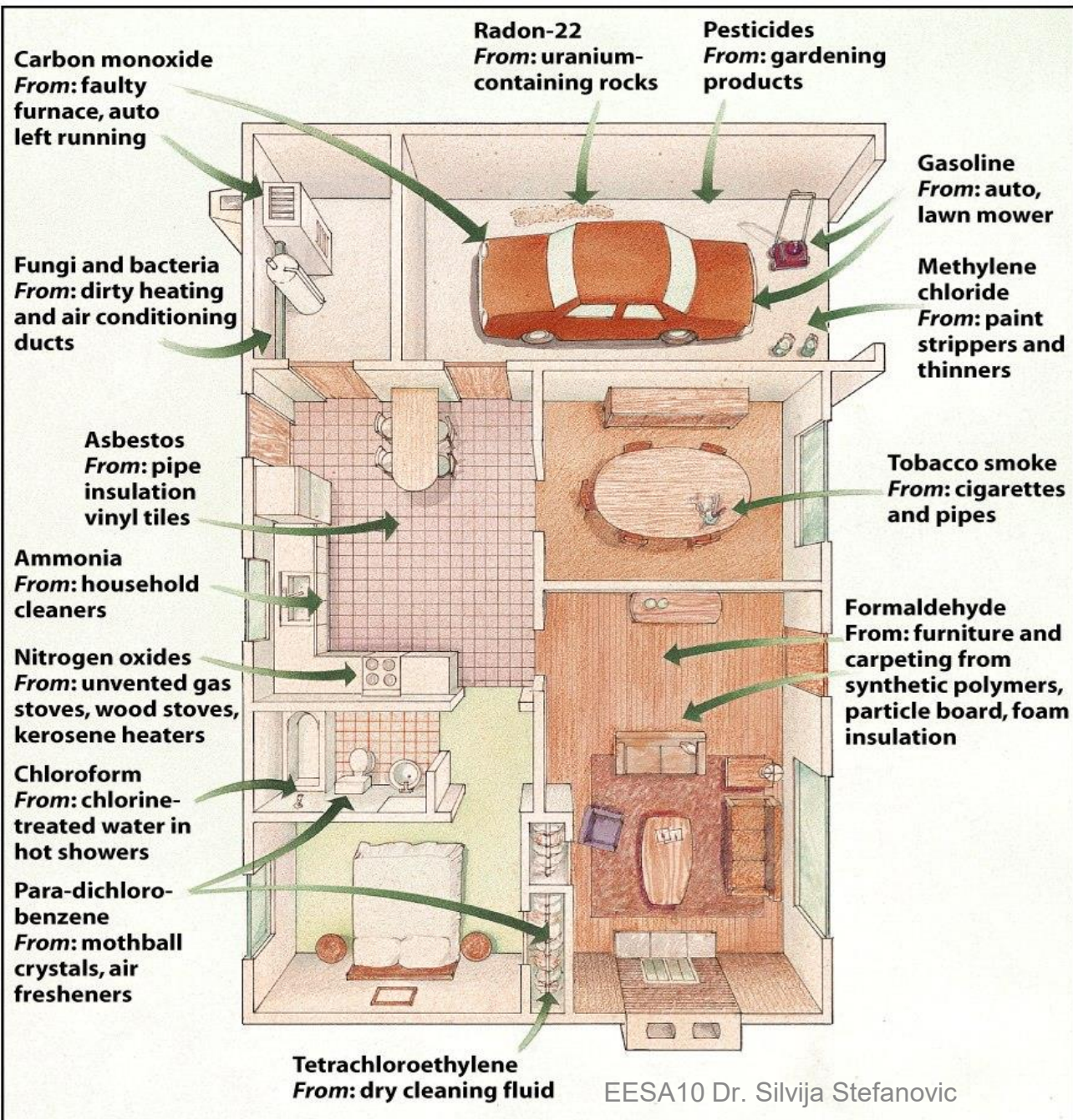
Sources of Pollutants

- Building materials and furnishing
- Asbestos insulation
- Wet or damp carpet
- Furniture made of certain pressed wood products
- Cleaning products and air fresheners
- Personal care and hobbies

Sources of Pollutants

- Pesticides
- Cooking
- Bathing
- Heating (combustion of oil, coal, wood)
- Radon
- Smoking
- Outdoor pollution



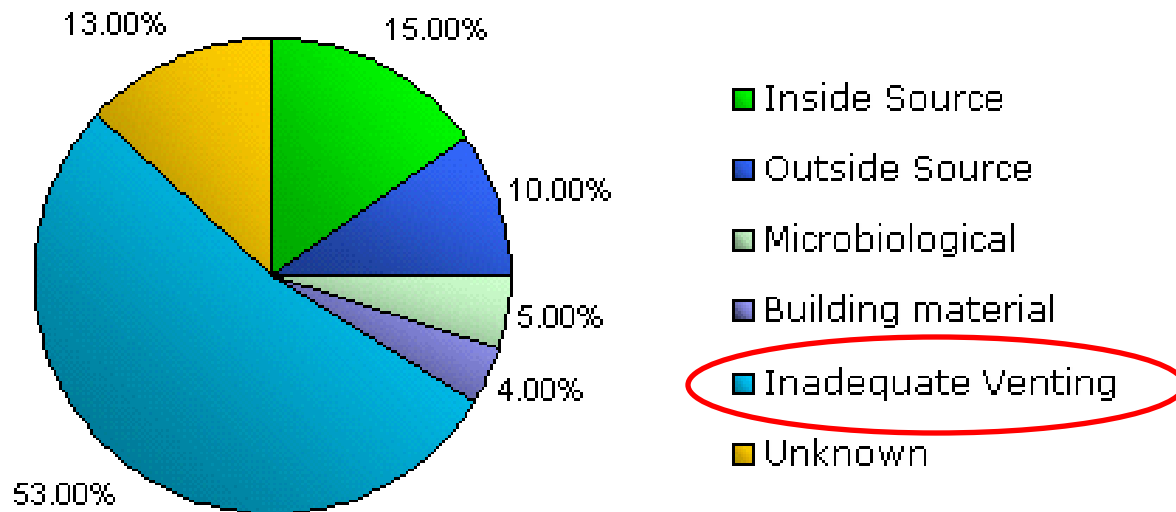


Unhealthy Construction Materials and Sick Buildings

- “Sick building syndrome”
 - Nonspecific symptoms experienced by occupants of a building
- “Sick building” designation
 - A building whose occupants experience such symptoms
- Building-related illness
 - Specific diagnosable illness, linked to specific feature of building

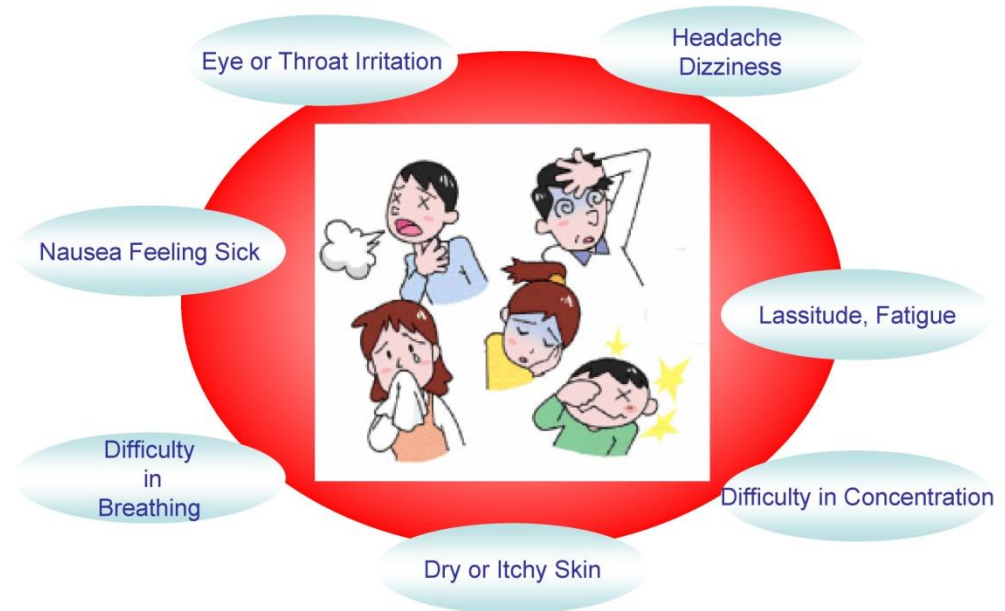
Sick Buildings Syndrome

NIOSH Indoor Air Investigations
Probable Causes of Sick Buildings



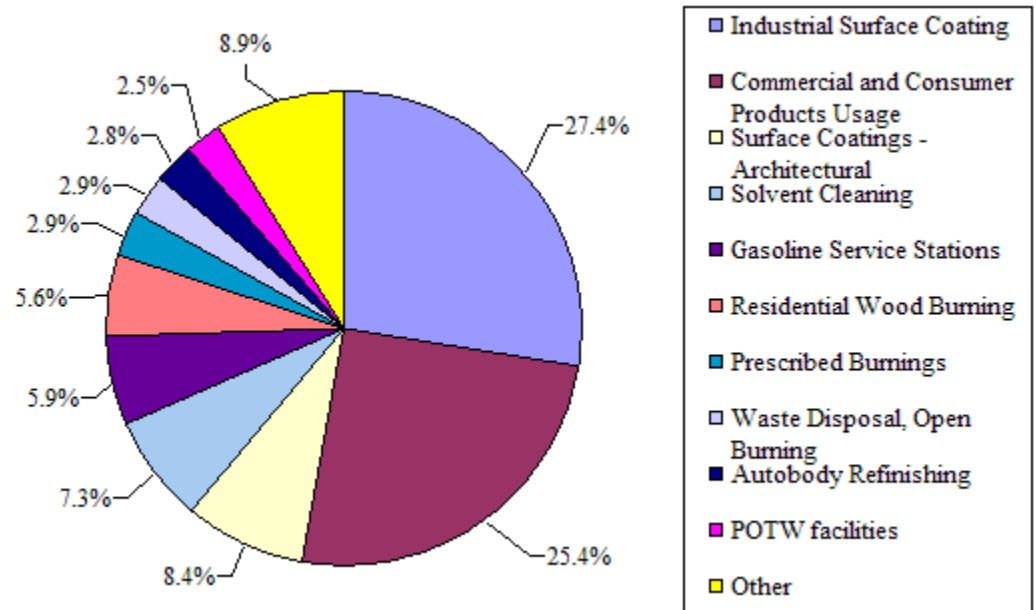
Health effects of Indoor Air Pollution

- Hard to detect by our senses
- Symptoms are similar, need years to develop
- Headaches, tiredness, dizziness, nausea, itchy nose, scratchy throat
- Asthma
- Cancer



Six Common Indoor Air Pollutants

- Asbestos
- Formaldehyde
- Mold and Moisture
- Secondhand Smoke
- Radon Gas
- Air Dust



Asbestos

- Group of six different fibrous minerals
- Have separable, long, strong and flexible heating resistant fibers



Fibrous **Asbestos**

Asbestos in the Environment

- Do not evaporate into air or dissolve in water, do not break down
- Fibers and particles may remain suspended in the air and carried long distances
- Not able to move through soil

Asbestos

- Used in:
 - Building materials (roofing shingles, ceilings and floor tiles, paper products, and asbestos cement products)
 - Friction products (automobile brakes and transmission parts)
 - Heat resistant fabrics, packaging and coatings



Asbestos paper wrap

Canada implemented a comprehensive ban on the import, manufacture, sale, and use of asbestos and asbestos-containing products, effective December 30, 2018

The ban prohibits new asbestos products from entering the market but requires awareness for older materials still found in buildings, necessitating proper handling and removal.

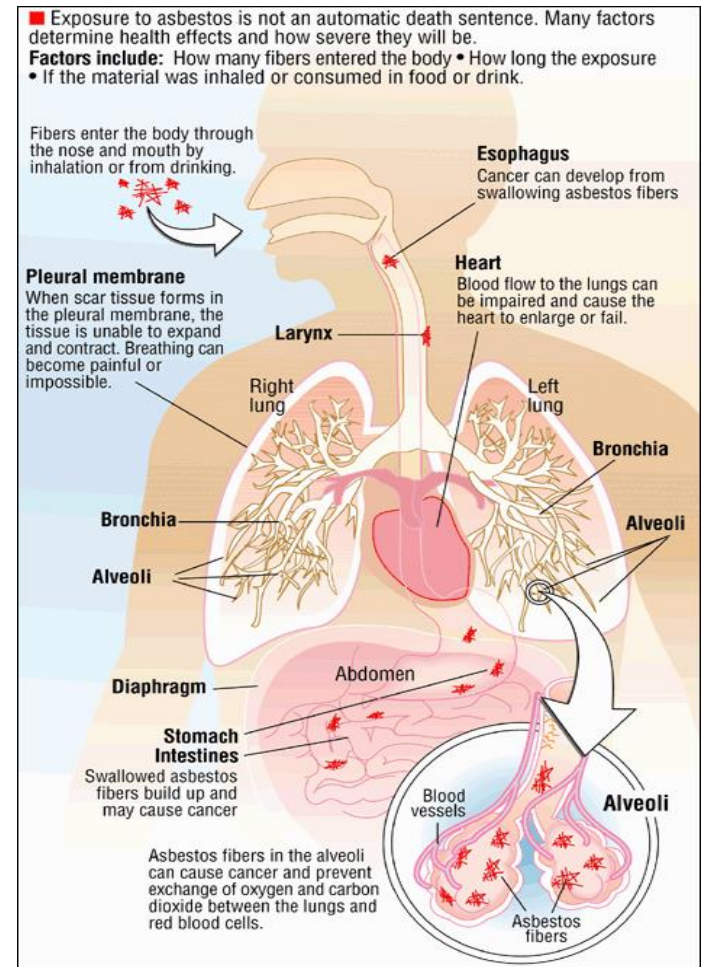
How might you be exposed to asbestos?

- In industrialized countries, nearly everyone has asbestos fibers in their lungs
 - Some risk of cancer
- People working in some industries
- People living near these industries
- During demolition work and remodeling
- From drinking water
(natural sources or asbestos containing cement pipes)



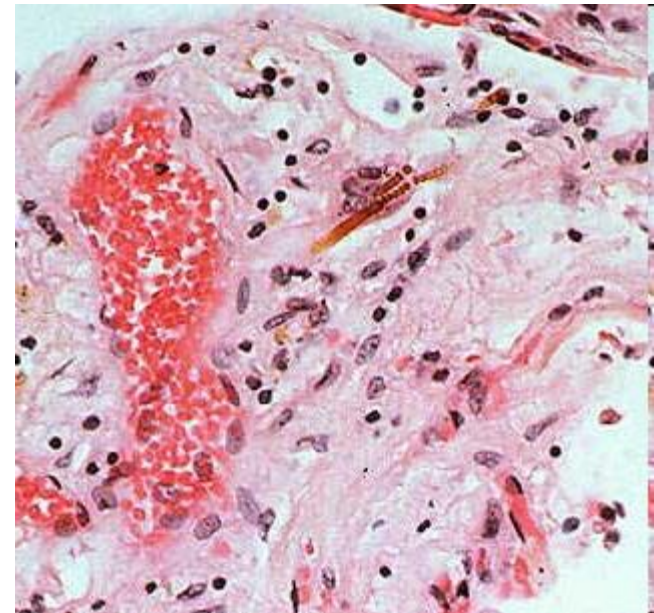
Health Effects of Asbestos

- Affect the lungs and the membrane that surrounds the lungs
- **Asbestosis**- Scar-like tissue, not in general public
 - difficulty breathing,
 - often cough,
 - heart enlargement
 - lead to disability and death



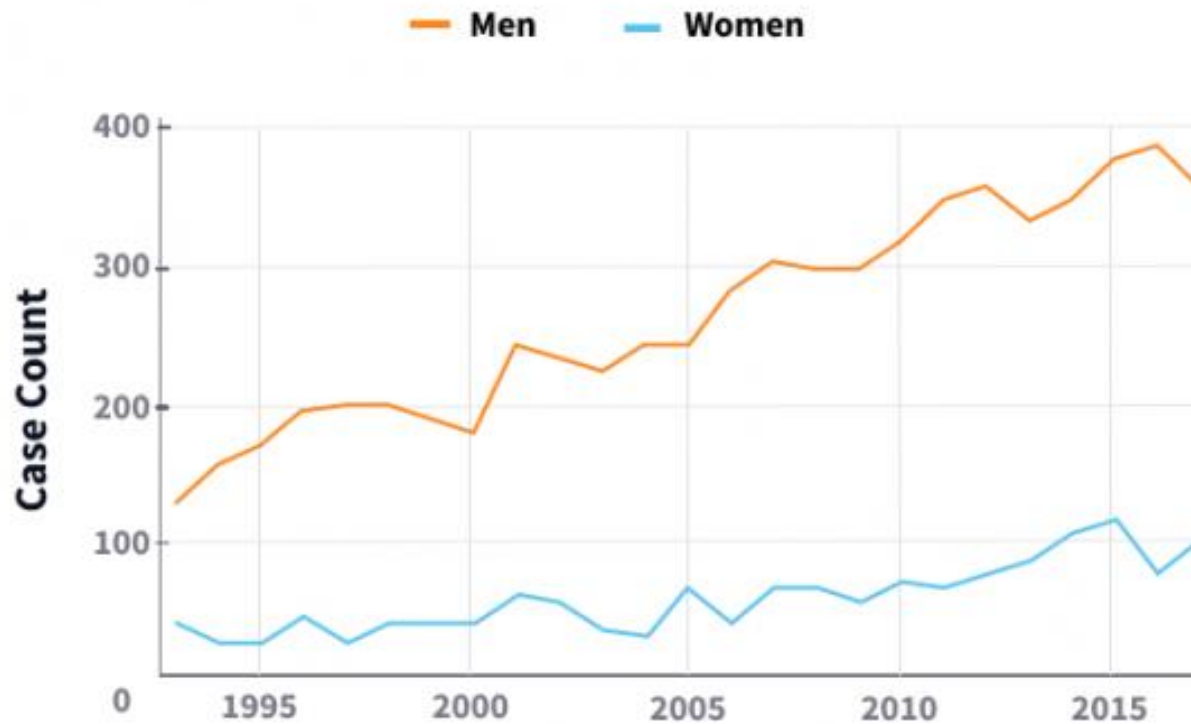
Health Effects of Asbestos

- **Plaques in the pleural membranes**
 - Thickened areas of tissue on the lung lining
- **Lung cancer, mesothelioma**
- Increase risk of getting other types of cancer (stomach, esophagus, pancreas, kidney)
- Risk increases with smoking
- Not related with birth defects
- Low levels can be measured in urine, faces, and mucus



Asbestos fibers lodged in the lungs.
Asbestos-related conditions affect the lungs and surrounding tissues

Steady Rise of Mesothelioma Cases in Canada



Source: Statistics Canada, 2022

Asbestos.com

Formaldehyde

- Volatile organic compound (VOC), naturally occurring gas, colourless, and strong smell
- Becomes a gas at normal room temperature
- Also released by burning wood and natural gas, by automobile and by cigarettes



Formaldehyde

- Glue or adhesives in pressed wood products (particleboards, MDF, plywood)
- Preservatives in some paints and cosmetics
- Coatings that provide permanent press quality to fabrics and draperies
- Finish used to coat paper products
- Certain insulation materials



**Fabrics and
Draperies.**

Health Effects of Formaldehyde

- Allergic reactions
 - Watery eyes, burning sensation in the eyes, nose and throat
 - Skin rashes
- Nausea
- Coughing
- Chest tightness
- Asthmatic reactions
- Cancer
- Some people very sensitive



Mold and Moisture

- Need moisture, does not need standing water, just requires high relative air humidity
 - Bathrooms and kitchens
 - Gym areas
 - Locker rooms
 - Leaky roof areas
 - Damp basements
 - On or within wood, paper, carpet and foods



Mold Health Effects

- Major source of indoor allergens
- Trigger asthma
- Produce Toxins
- Produce Irritants



Water Damage & **Mold** Growth



Mold under microscope⁴⁶

Mold and Moisture

- The way to control indoor mold growth is to control moisture (maintaining the relative humidity between 30-60%)
- Often undiscovered
- Produce tiny spores
- Discoloration and odour problems

Second-hand Smoke

- Contain 4 000 compounds (CO and Formaldehyde), 40 are carcinogens
- A non-smoker exposed to secondhand smoke has a 25% increased chance of developing lung cancer compared with not exposed non-smoker
- EPA estimates that each year 3 000 lung cancer deaths among nonsmoking adults in US
- Health Canada estimates that more than 300 non-smokers die from lung cancer each year because of such exposure



Second-hand Smoke

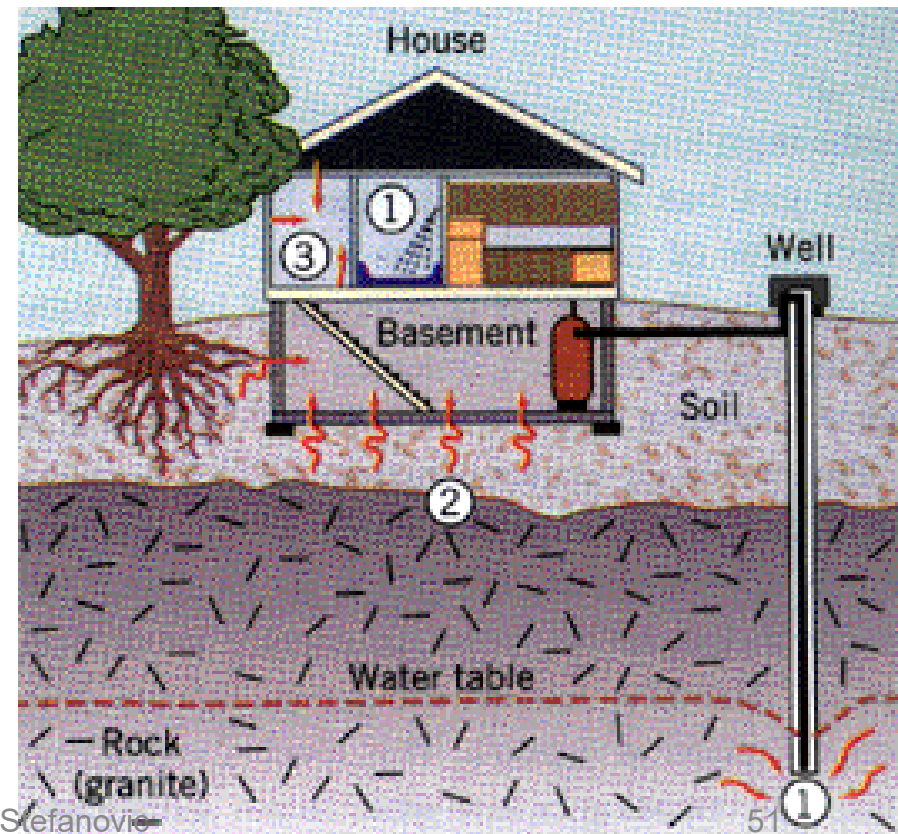
- **Environmental tobacco smoke** (secondhand smoke):
 - Adults: heart disease, heart attack, lung cancer, hearing loss, eyes, nose and throat irritation
 - Children: SIDS, asthma, pneumonia, bronchitis, ear infection and hearing loss
- Prevalence is generally falling in most countries worldwide, including many low- and middle-income countries (where 80% of the world's smokers live)
- Rates are stable or decreasing, while the absolute number of smokers is not declining, and some countries still report rising numbers.
 - More than 8 million people projected to die from smoking-related causes by 2030

Third-hand Smoke????



Radon Gas

- Colorless, odorless, tasteless
- Naturally occurring
- Radioactive decay of uranium
- From soil and rock into basements and lower floors indoor air
- Dissolved in groundwater, pumped into wells and then into homes
- In construction building blocks



Radon Gas in Buildings

- Natural hazards in some regions
 - Begins series of rapid breakdowns
 - Radon and some progeny are alpha emitters; lung cancer risk
 - Often simple to detect and remediate



Radon Gas Detector

Air Dust

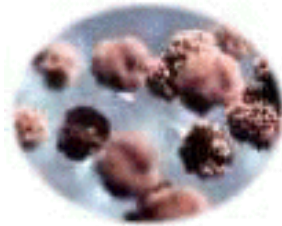
- Heating and cooling -forced air system
- Dust particles
- Pollen or other debris
- Duct Cleaning Service Providers



Dust Mites



Pollen



Mold



Pet Dander



Bacteria / Viruses